

Charan Vardham

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EDUCATION

University of Wisconsin Madison

Madison, WI

Bachelor of Science in Computer Science, Minor in Mathematics - 3.9 GPA (CWRU transfer) Aug. 2026 – April 2028

- **Relevant Courses:** Data Structures, Algorithms, Discrete Mathematics, Linear Algebra, Computer Security

EXPERIENCE

Open Source Contributor

May 2026 – Present

PyTorch

Remote

- Fixed **3** bugs in PyTorch Inductor's `torch.compile()`, achieving a **2x** improvement in graph compilation speed
- Added CPython's `descr_check` validation to Dynamo's C-descriptor variable tracking, fixing incorrect control flow during tracing when incompatible types bind to C descriptors

Undergraduate Student Researcher

Aug 2025 – Present

Case Western Reserve University

Cleveland, OH

- Designed and trained deep learning CNN models in PyTorch on **5000+** CT scan slices for cross-kernel image reconstruction, optimizing model accuracy using PSNR and SSIM evaluation metrics
- Outperformed state-of-the-art image translation baselines (Pix2Pix, CycleGAN, CUT) by up to **54%** lower FID and **10%** higher PSNR
- Submitted manuscript to the Image Processing track of SPIE Medical Imaging 2027, an internationally recognized venue for medical imaging research

xLab Intern

Jan 2026 – May 2026

Weatherhead School of Management, CWRU

Cleveland, OH

- Built an AI case study platform using LLMs for business simulation, serving **200+** students across **3** MBA courses
- Deployed a full-stack application with Next.js (TypeScript), Tailwind CSS, PostgreSQL, and Prisma

AI/ML Intern

May 2025 – Aug 2025

IQVIA

Remote

- Built a scalable RAG-based medical chatbot processing 200MB+ of clinical documents to reduce hallucinations and improve factual accuracy
- Implemented document ingestion, text chunking, embedding generation, and retrieval pipelines achieving **90%** retrieval accuracy with MongoDB-backed chat history
- Worked on a separate voice-enabled chatbot project, integrating NVIDIA Riva APIs for real-time STT/TTS with an LLM backend

PROJECTS

GPT-2 Pretraining ([link](#)) | *PyTorch, Python, CUDA, SLURM*

February 2026

- Trained GPT-2 (**124M** parameters) from scratch on **10B** tokens using NVIDIA H100 GPUs, achieving final loss of **3.0** with a throughput of **260K** tokens/sec
- Implemented a custom **FlashAttention-2** kernel in **Triton** and Fully Sharded Data Parallel (FSDP)

Real-Time Credit Card Fraud Detection ([link](#)) | *Python, TypeScript, XGBoost, AWS, Docker*

June 2026

- Built an end-to-end ML fraud detection system using a stacking ensemble model trained on the IEEE-CIS dataset and deployed in a Docker container on ECS Fargate via SST
- Implemented a Next.js 15 inference API (predict, transaction history, health check) with TypeScript types shared across the stack, writing predictions to DynamoDB on every request
- Designed CI/CD on GitHub Actions with TypeScript type-checking, Python linting, and automated SST deployment to AWS

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL, R, HTML/CSS, C++

Frameworks / Libraries: React, Node.js, Next.js, FastAPI, Material-UI, Pandas, NumPy, Matplotlib, Scikit-Learn, PyTorch, TensorFlow, PyTorch Lightning, Hugging Face Transformers

Developer Tools / Deployment: Git, Docker, VS Code, IntelliJ, PyCharm, Visual Studio, REST APIs, AWS, CI/CD, Jenkins, Kubernetes, Unix

Testing / ML Skills: Unit testing, Data preprocessing, Feature engineering, Model evaluation metrics (PSNR, SSIM), RAG pipelines, LLM integration